PRODUCT REQUIREMENTS DOCUMENT

College Transportation Management Website

Scope: Common Login Page and Student Dashboard

Document Detail	Value
Version	1.1 – Student Dashboard Scope
Date	1 November 2026
Project Type	Web-Based College Transportation Management System

1. Project Overview

The College Transportation Management Website is a centralized web-based platform for managing college transportation. This document covers only the common login page and the Student Dashboard. Teacher, Driver, and Admin dashboard requirements will be documented separately in a later phase.

The Student Dashboard will allow students to view their registered details, confirm whether they intend to board their bus, receive transport notifications, submit complaints, and track their assigned bus on a live map.

2. Common Login Page

The website will provide one common login page for registered users.

2.1 Login Fields

- Username / User ID
- Password
- Login button

2.2 Role-Based Redirection

After successful authentication, the backend will identify the user's role and direct the user to the appropriate dashboard. This PRD defines only the Student Dashboard; the other dashboards will be specified separately.

- Verify the submitted login credentials.
- Identify the account role using the registered user record.
- Redirect students to the Student Dashboard.
- Restrict access to pages and data that the logged-in user is not authorized to view.

3. Student Dashboard – Overview

The Student Dashboard is the main interface for students to access their transportation information and services.

Dashboard layout based on the supplied sketch:

- Top area: Student Dashboard heading and notification icon/box.
- Left section: Student details.
- Centre section: Student check-in status and automatic checkout information.
- Right section: Student photo, name and USN, with a complaint form below.
- Lower section: Live bus tracking map, bus details and route information.

4. Student Details

Location: Left side of the Student Dashboard.

Display the student's registered details:

- Student Name
- USN (University Seat Number)
- Class
- Address
- Department
- Phone Number
- Email ID

Requirements:

- Retrieve details from the student record in the database.
- Display the information in a clear, organized panel.
- Students can view their details; changes are made only by authorized users.

5. Student Check-In

Purpose: Let the student notify the transport system that they intend to board the bus. This helps the driver know which students are waiting at which stops.

5.1 Check-In Rules

- A student must check in at least 10 minutes before the scheduled bus arrival time at their assigned stop.
- The check-in option closes 10 minutes before the scheduled arrival time.
- The system must reject a check-in submitted after the cutoff.
- A successful check-in must be saved against the student and the relevant bus trip.
- The check-in information must be made available to the transport/driver workflow.

5.2 Status Indicators

Indicator	Status	Meaning
Green	Checked In	The student has confirmed their intention to board.
Red	Not Checked In	The student has not confirmed boarding.

Example: If the bus is scheduled to arrive at the student's stop at 8:00 AM, check-in closes at 7:50 AM. The student must check in before 7:50 AM.

Decision to confirm: The 10-minute cutoff should be based on the scheduled arrival time at each student's stop, not a single general bus departure time.

6. Automatic Check-Out

Purpose: Automatically record when students reach the college.

- When the bus reaches the college, the system should identify students associated with that active trip.
- The system should automatically update their trip/check-out records.
- Check-in and check-out timestamps should be retained as travel history.

Technical note: Use a configured GPS geofence around the college to detect arrival. The system should avoid checking out students based only on an approximate map location.

7. Notifications

Location: Near the top of the Student Dashboard, above the student profile area.

Display transport-related announcements and updates published by the administrator.

- Bus delays.
- Route changes.
- Bus schedule changes.
- Driver or bus-related updates.
- General transportation announcements.

Requirements:

- Show the latest notifications first.
- Make important updates easy to notice.
- Display the notification date/time where appropriate.

8. Student Profile

Location: Right side of the Student Dashboard.

Display the student's profile photograph, name, and USN. The information should correspond to the currently logged-in student.

9. Complaint Form

Purpose: Allow students to submit transportation-related complaints.

9.1 Form Fields

- Complaint date (automatically generated)
- Complaint description
- Submit button

9.2 Requirements

- Validate that the complaint description is not empty.
- Save the complaint in the database with the student's ID and submission date/time.
- Make the complaint available to the authorized administrator for review.

Complaint status tracking (Submitted, Under Review, Resolved) can be included in a later iteration if required.

10. Live Bus Tracking – GPS Map

Purpose: Allow students to see the current location of their assigned bus and follow its movement on a map.

10.1 Information Displayed

- Current bus location using a moving bus marker
- Bus number
- Assigned driver details
- Assigned bus route
- Bus movement along the route

10.2 GPS and Map Requirements

- The system must receive GPS coordinates from a bus GPS device or an authorized driver's mobile application.

- The latest bus location should be refreshed approximately every 10 seconds.
- The map should move/update the bus marker as new coordinates arrive.
- The assigned route should be visible so the student can understand the bus journey.
- Students should only see the bus assigned to them or otherwise authorized for their trip.
- Show the time of the last location update so students can judge whether the location is current.

GPS refresh target: Every 10 seconds.

Important dependency: A website cannot determine a moving bus's location by itself. A GPS-enabled device or a driver's phone must send location updates to the backend. If GPS data becomes unavailable, the interface should show the last update time and indicate that the location may be out of date.

11. Student Dashboard Workflows

11.1 Login

- Student opens the website.
- Student enters username and password.
- Backend verifies the credentials and student role.
- Student is redirected to the Student Dashboard.

11.2 Check-In

- Student views the assigned bus and stop schedule.
- Student selects Check In before the cutoff.
- Backend validates the deadline and records the check-in.
- The updated status becomes visible to the relevant transport workflow.

11.3 Live Tracking

- GPS device or authorized driver app sends coordinates.
- Backend receives and makes the latest location available.
- Student dashboard refreshes the map data.
- Bus marker updates approximately every 10 seconds.

11.4 Automatic Check-Out

- Bus enters the college geofence.
- System confirms the active trip and associated students.
- System records checkout for eligible students.
- Trip history is updated.

12. Non-Functional Requirements

Performance

- Student Dashboard should load efficiently.
- GPS location should refresh approximately every 10 seconds when data is available.

Security and Privacy

- Use secure authentication and password storage.
- Students should only access their own personal details and authorized bus information.
- Protect student and location data from unauthorized access.

Reliability

- Handle temporary GPS/network interruptions.
- Display the last successful GPS update time.
- Store check-in and checkout records reliably.

Usability

- Use a clear, simple dashboard layout.
- Use distinguishable check-in status indicators.
- Support desktop and mobile screen sizes.

13. Acceptance Criteria

- A student can log in through the common login page and reach the Student Dashboard.
- The dashboard displays the correct student's details and profile information.
- The student can check in before the 10-minute cutoff.
- The system rejects check-ins at or after the cutoff.
- The check-in status is saved and made available to the relevant transport workflow.
- The student can view transport notifications.
- The student can submit a complaint.
- The map displays the assigned bus, route, bus number and driver details where available.
- The bus marker refreshes approximately every 10 seconds when GPS updates are received.
- The system records checkout after confirmed arrival at the college geofence.

14. Out of Scope for This Document

The following modules are intentionally not specified in this version and will be documented later:

- Teacher Dashboard.
- Driver Dashboard.
- Admin Dashboard and management tools.
- Detailed database schema and API design.
- SMS/email notification integration.

End of Student Dashboard PRD